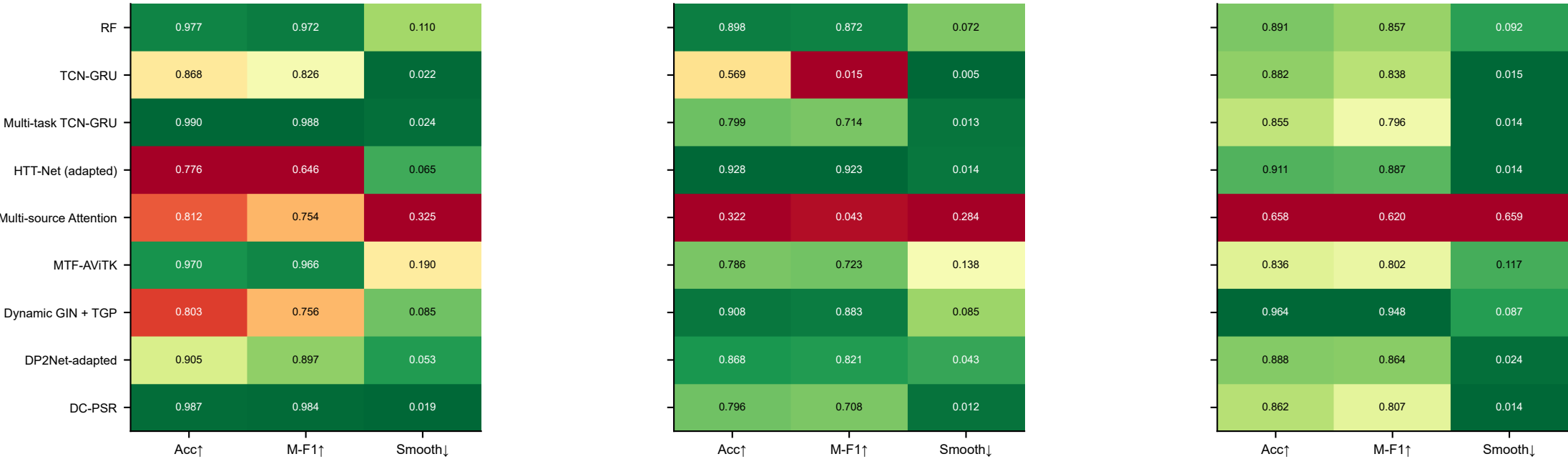
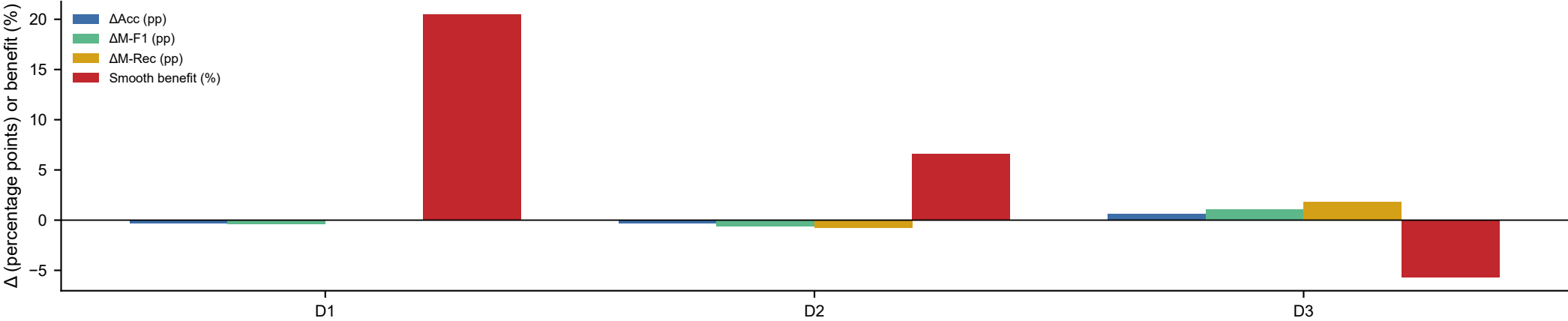


Cross-target-condition (PHM2010 D1/D2/D3) and cross-dataset (PHM2010/NASA Milling/MTW-CM) generalization: no single method dominates every task, but M-F1 and trajectory-consistency gains from DC-PSR stay directionally consistent

(a) Within-task normalized performance, D1/D2/D3 × 9 methods (color = within-task-and-metric normalized benefit; numbers = raw values)



(b) Multi-task TCN-GRU → DC-PSR paired delta, PHM D1/D2/D3 (Smooth benefit = Smooth_backbone - Smooth_DC-PSR, positive = improvement; raw Smooth is never itself flipped)



(c) Cross-dataset Multi-task TCN-GRU → DC-PSR (B11→B12) delta matrix (green = improvement in the metric's own direction, red = regression; color normalized per-column since columns mix pp and % units; raw ΔAcc/ΔM-F1 in pp, Smooth/Jump shown as % benefit)

